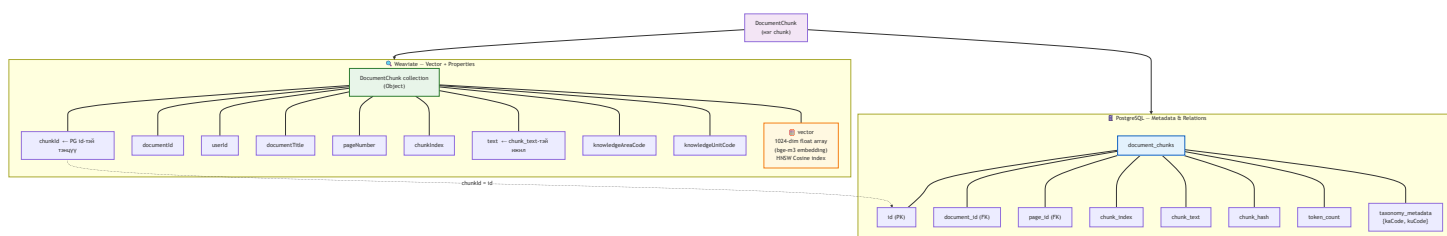


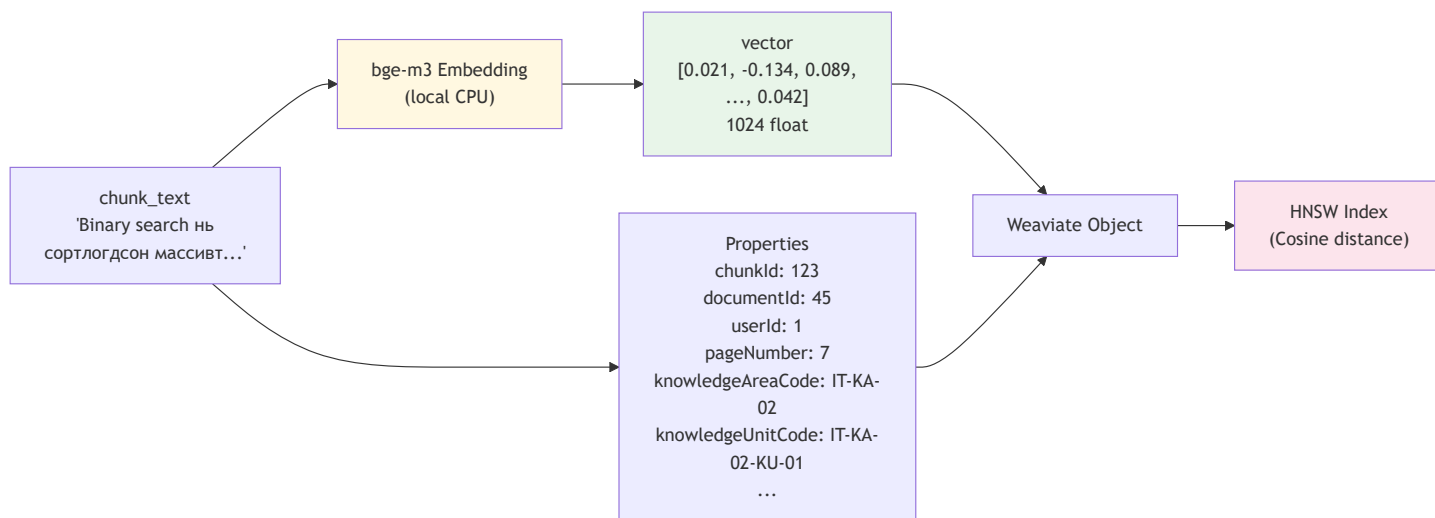
Smart Assistant for Learners — Системийн диаграммууд

0. Weaviate Vector DB — Chunk хадгалалтын бүтэц

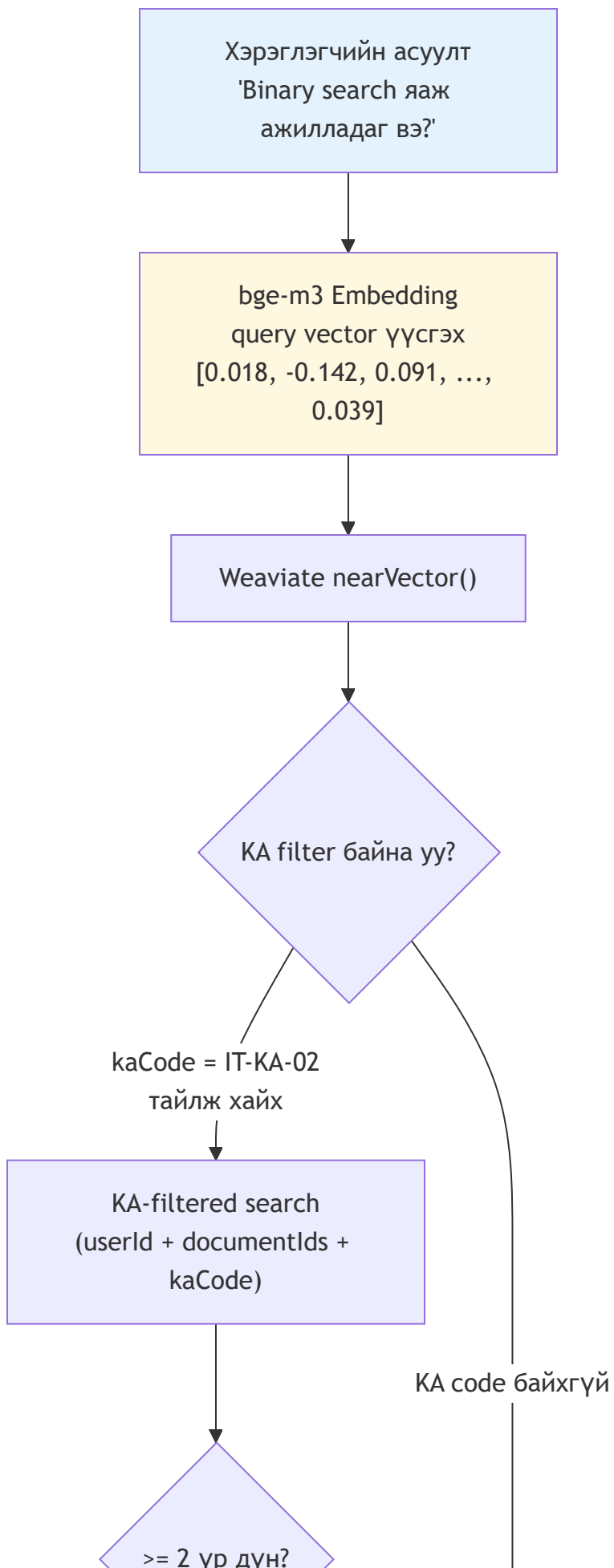
0.1 PostgreSQL vs Weaviate — Chunk өгөгдлийн хуваарилалт

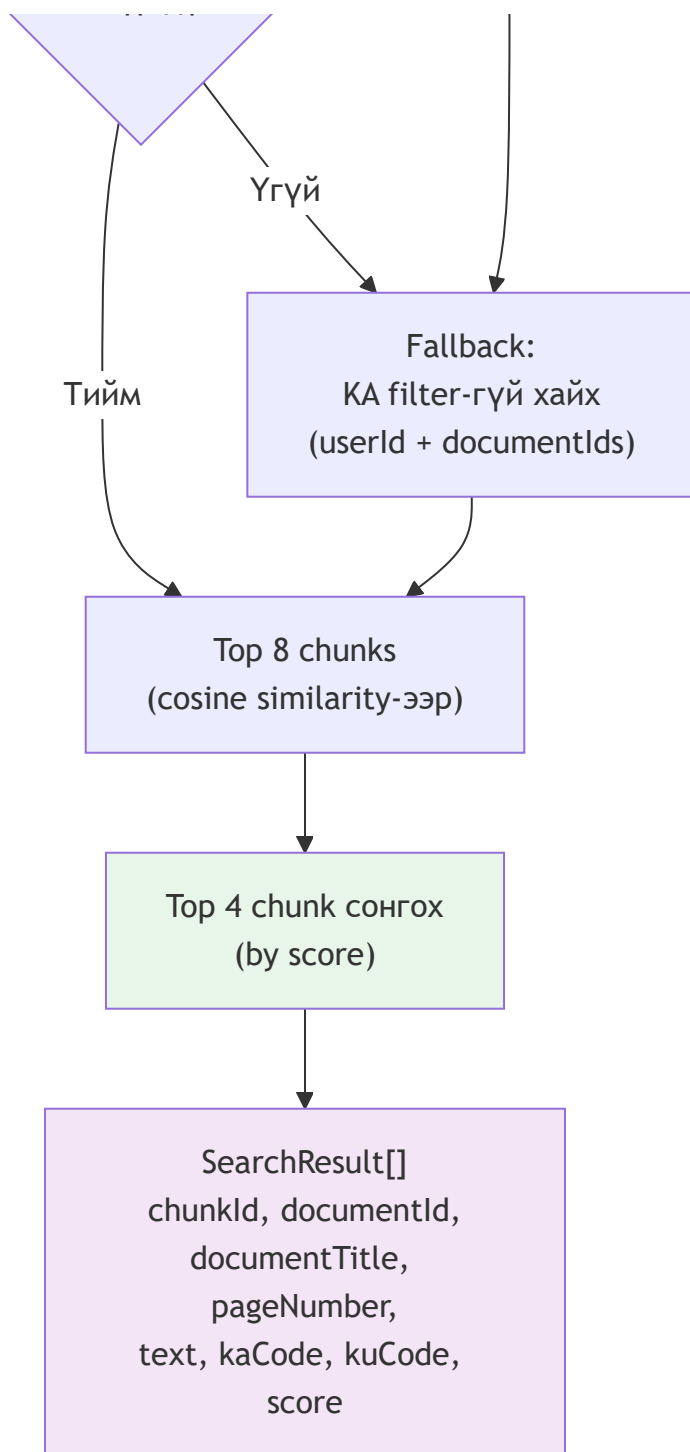


0.2 Chunk индексийн урсгал (Upload → Weaviate)



0.3 Semantic Search үрсгал (Query → Top-K chunks)

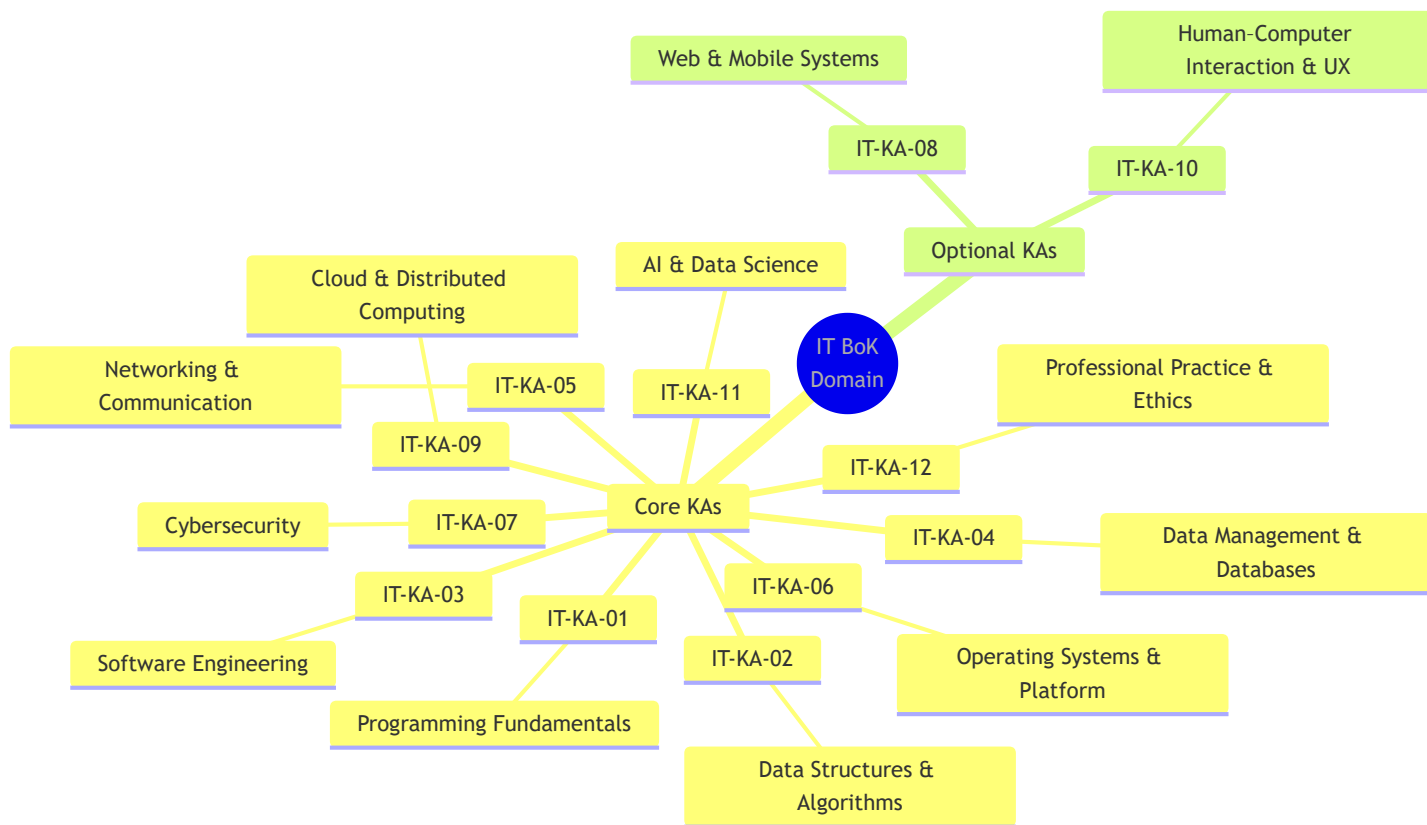




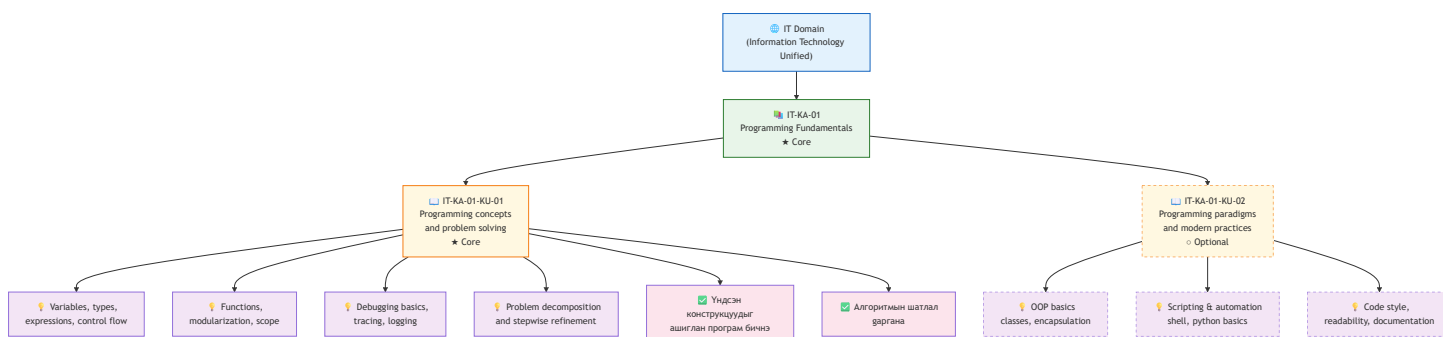
0. Тахоноту бүтцийн диаграмм

0.1 IT Body of Knowledge — Ерөнхий бүтэц (12 KA)

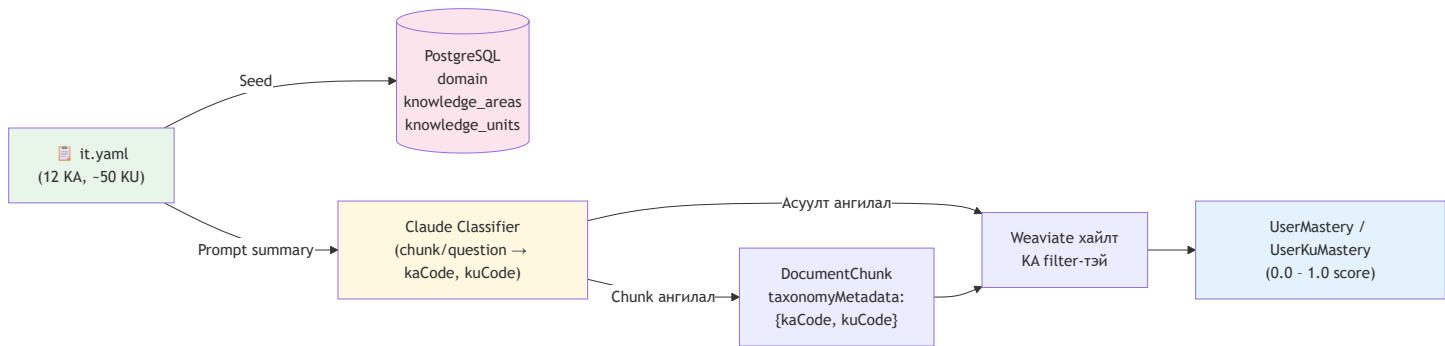
■ = Core (MVP-д заавал), ■ = Optional



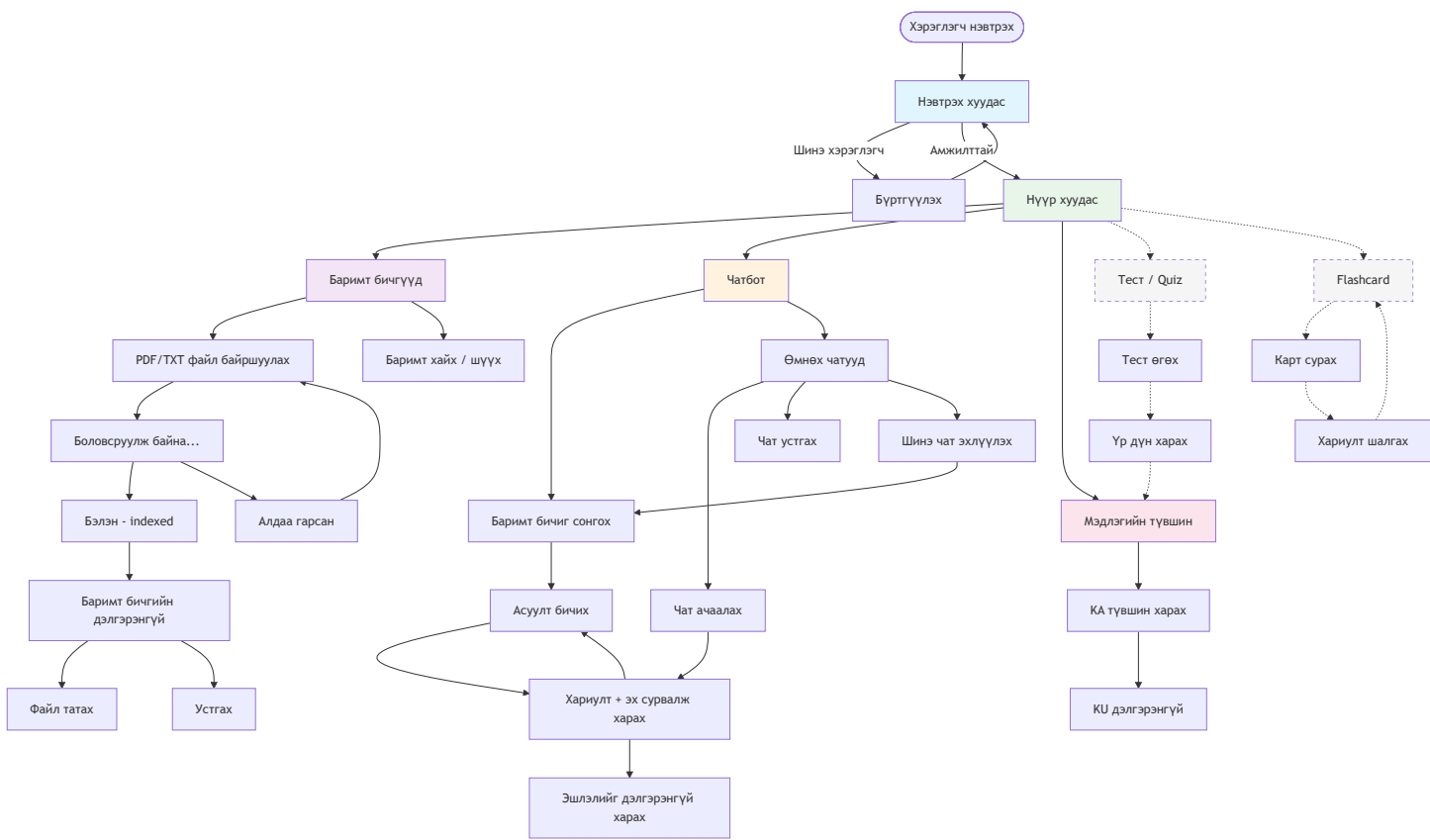
0.2 Тахому шатлалын бүтэц — Нэг КА-ийн жишээ (IT-KA-01)



0.3 Taxonomy → Системд хэрхэн ашиглагдах

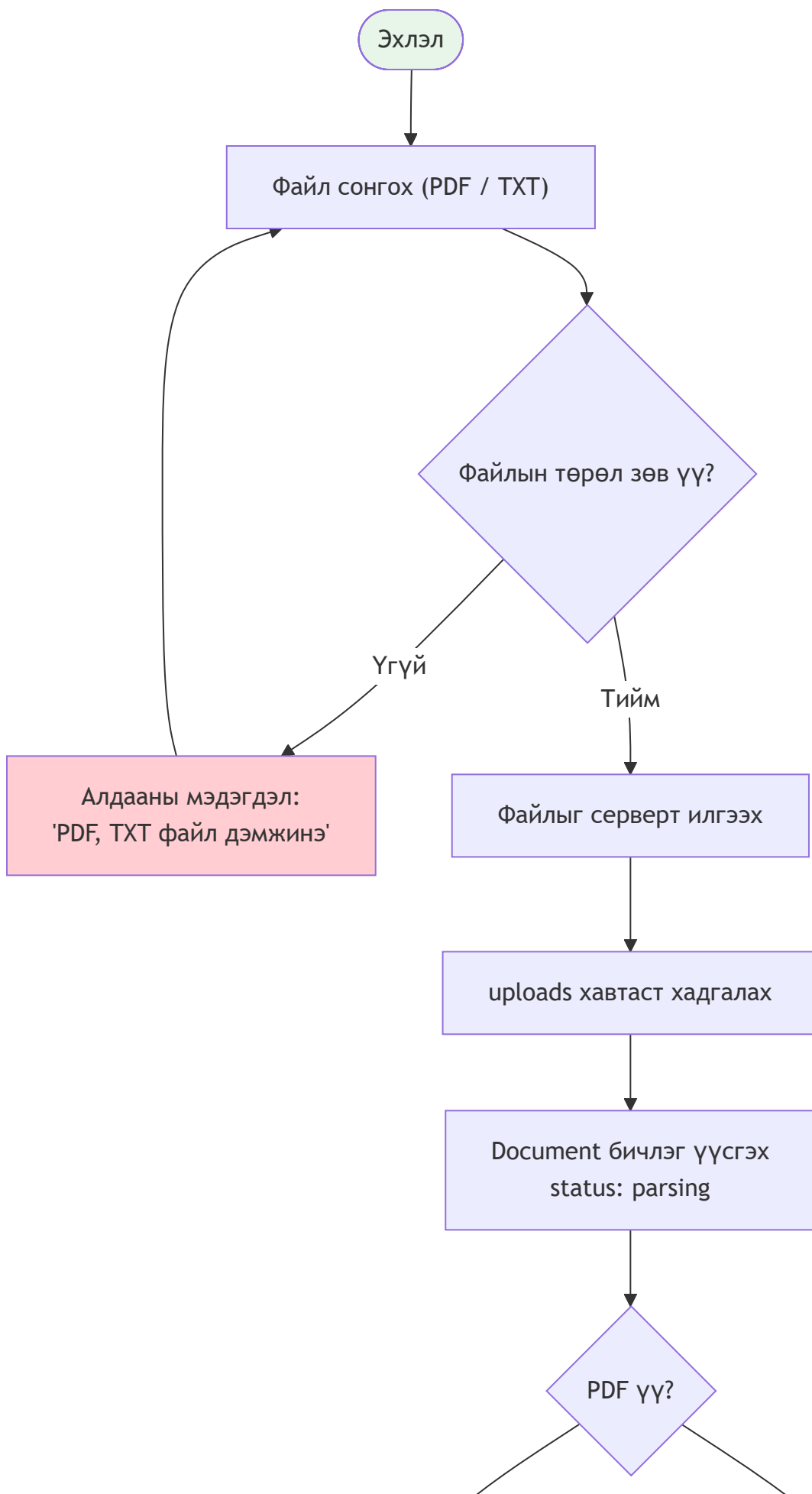


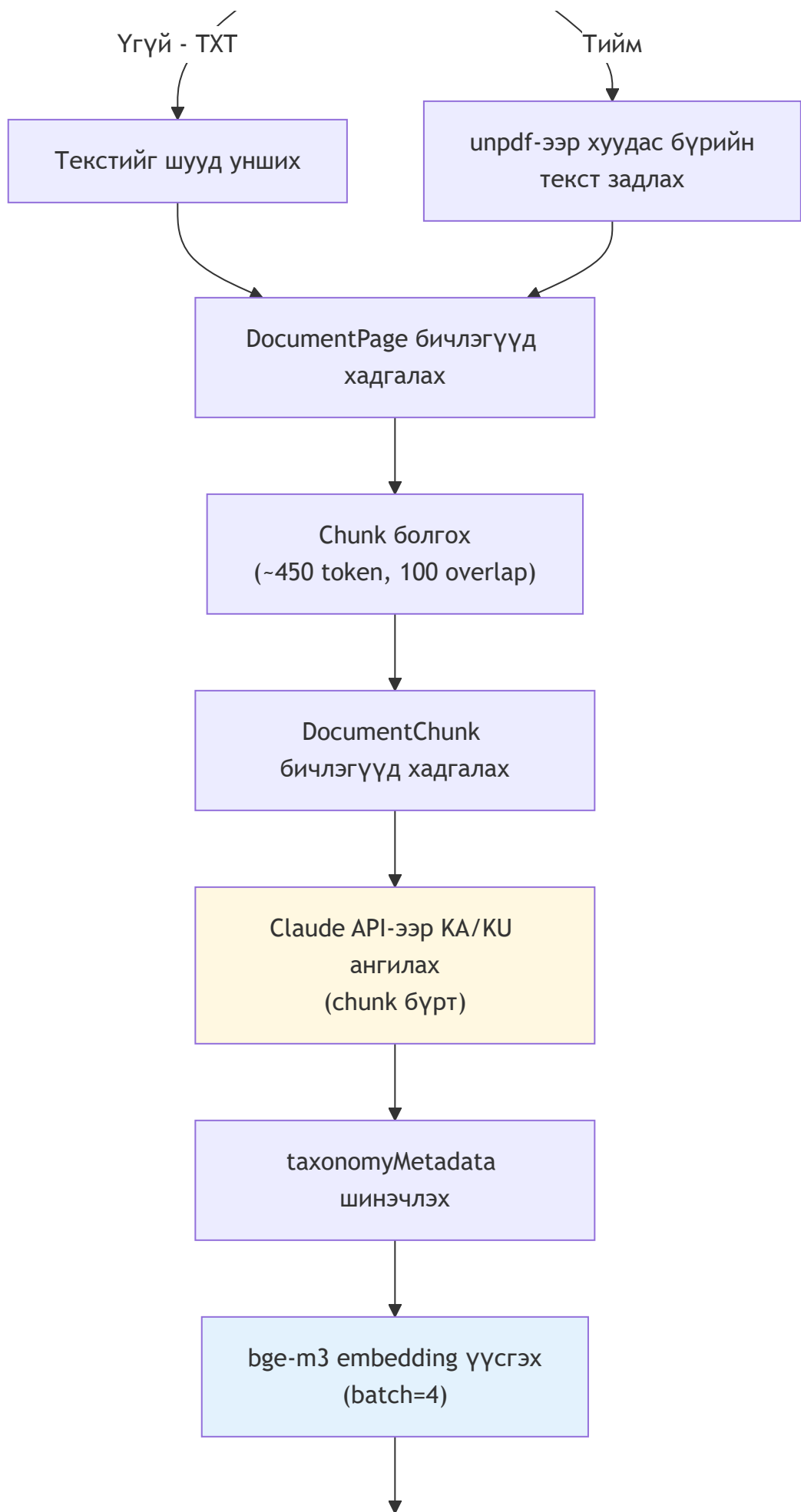
1. User Flow Diagram — Хэрэглэгчийн урсгал

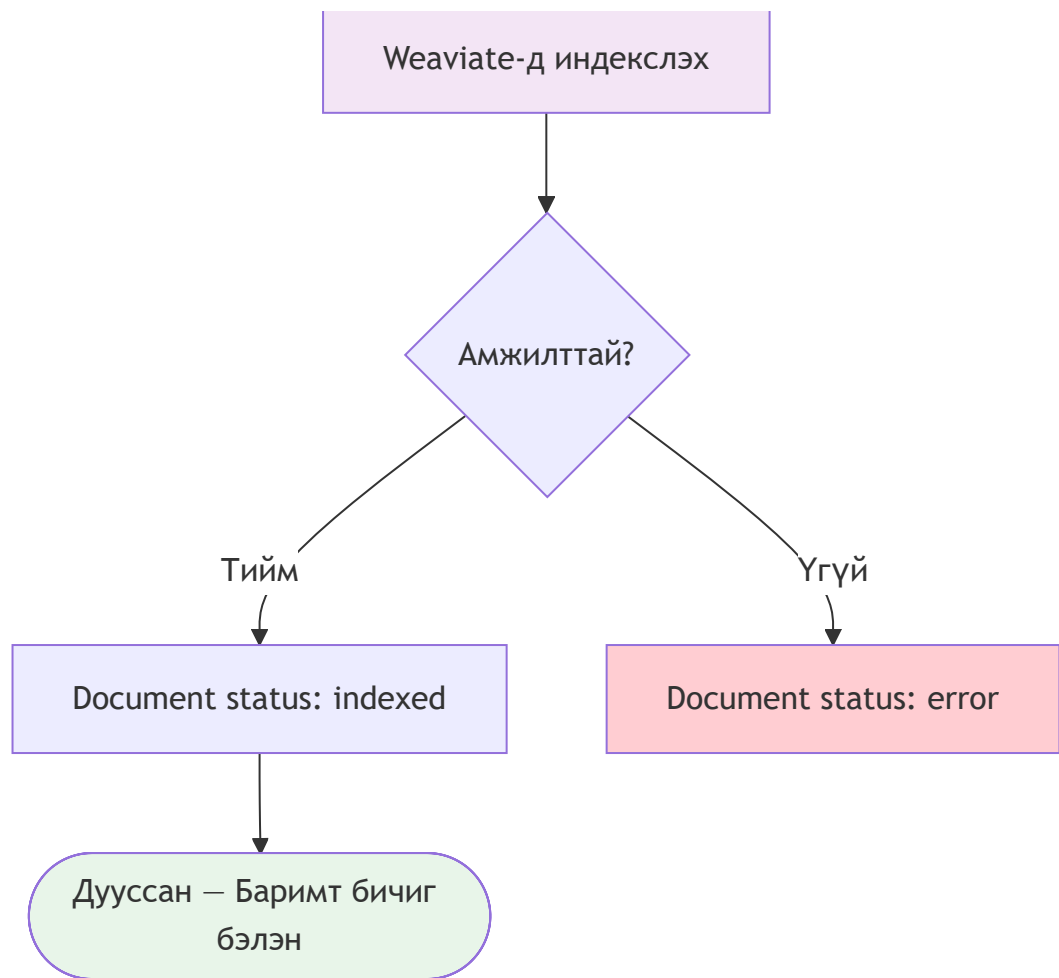


1.1 Document Upload Workflow — Баримт бичиг

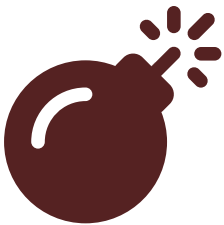
байршуулах урсгал





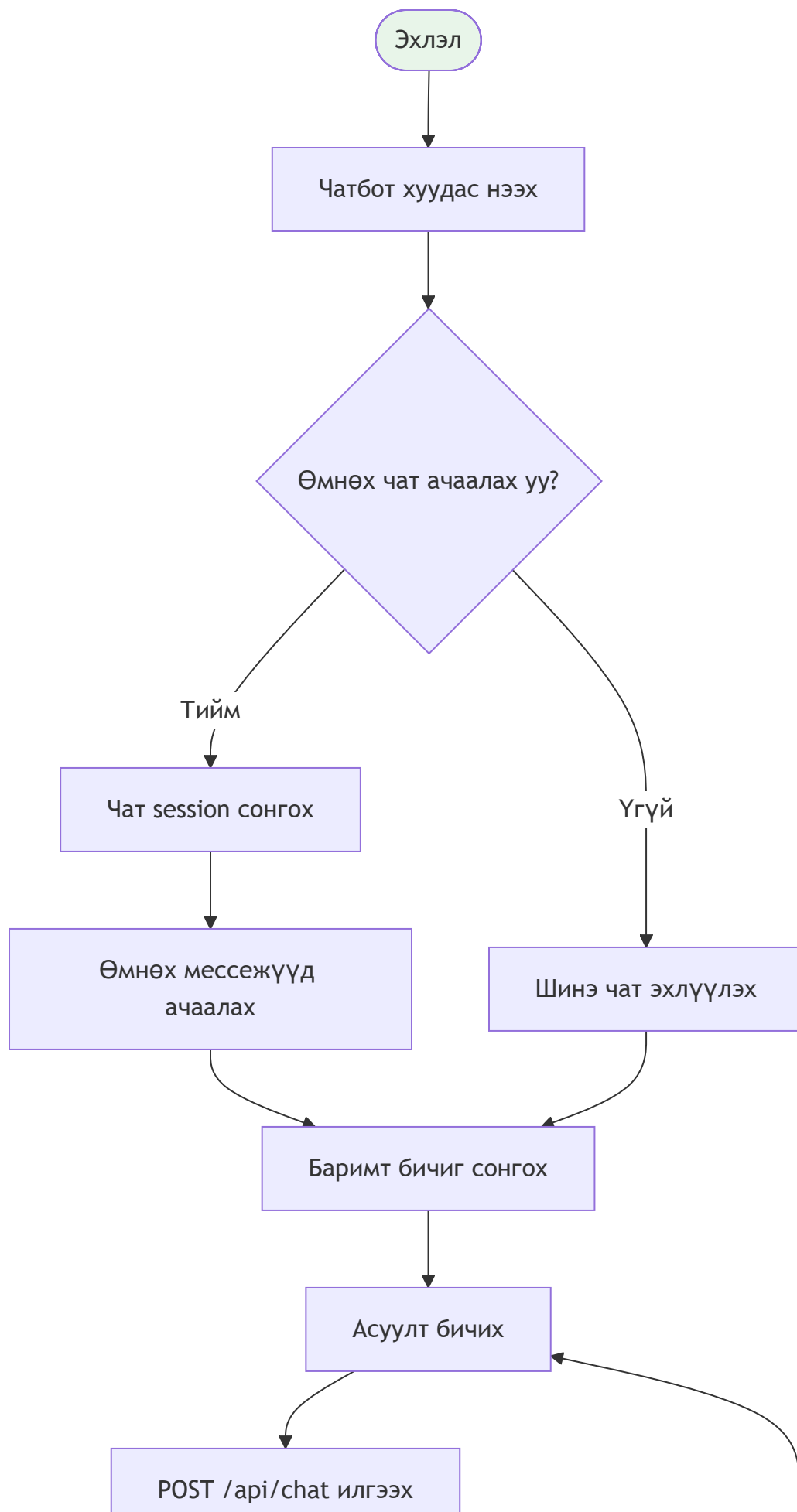


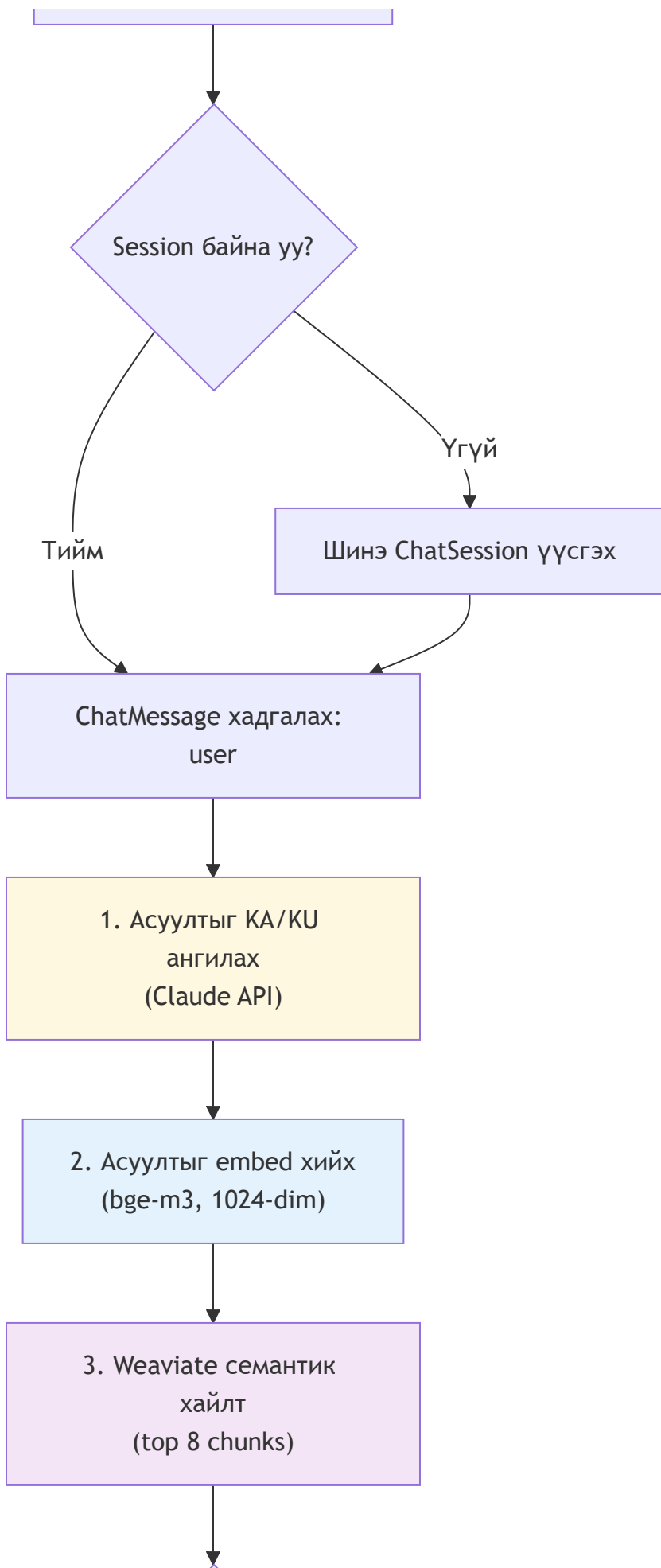
Зураг 5.2.1.2. Ingestion pipeline-ийн урсгал диаграмм

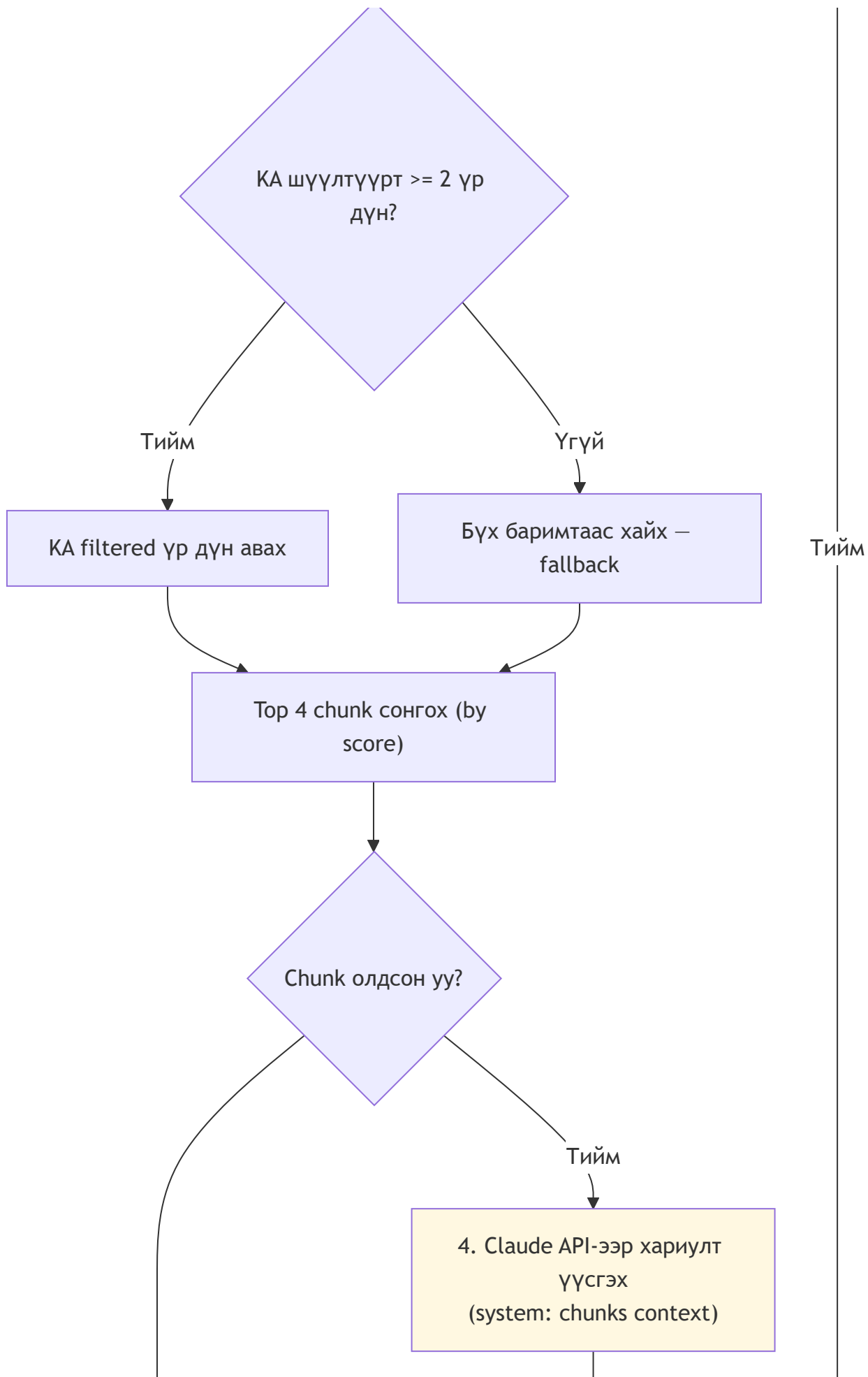


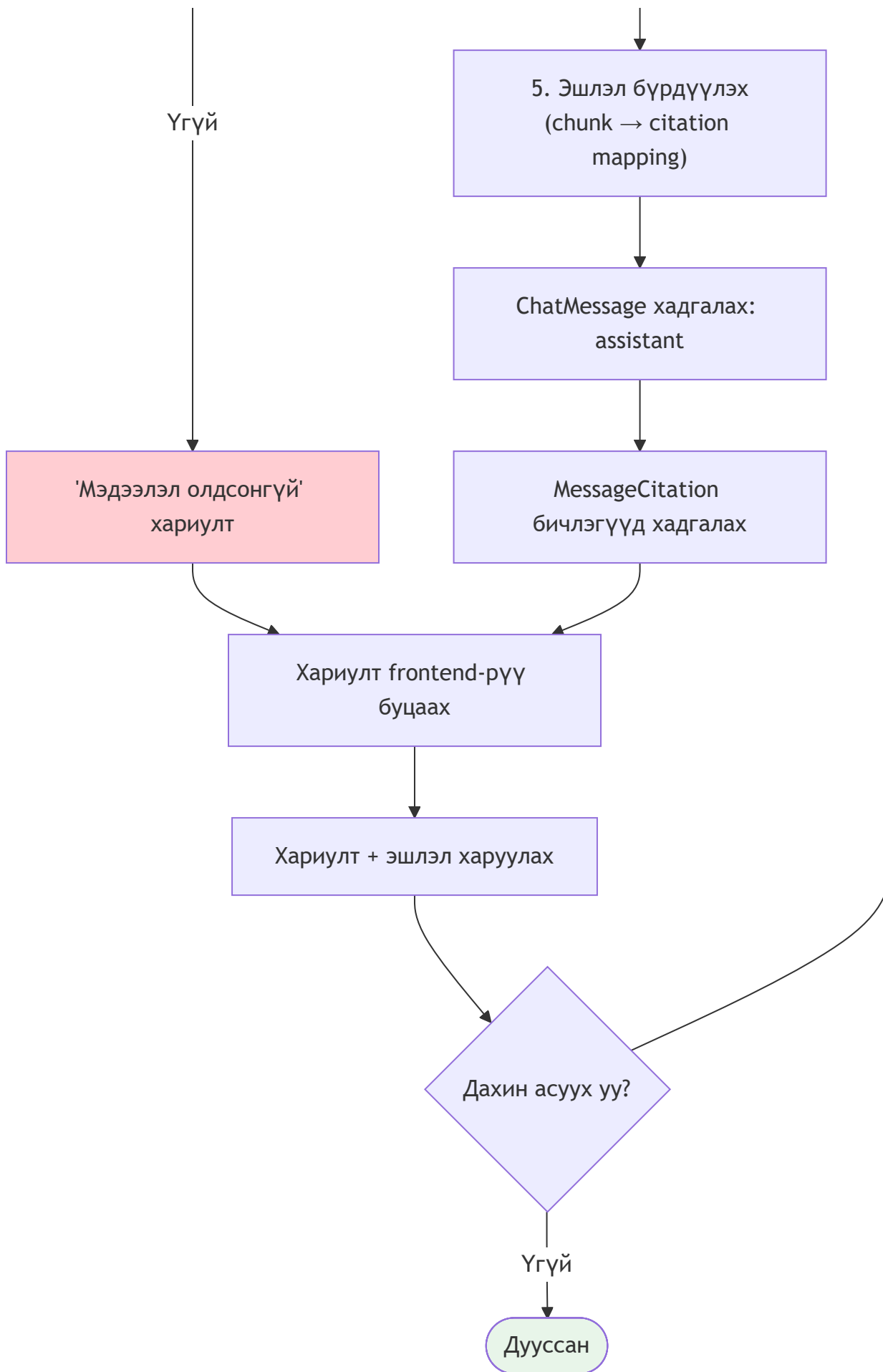
Syntax error in text
mermaid version 11.13.0

1.2 Chat Workflow — Чатбот асуулт-хариулт урсгал

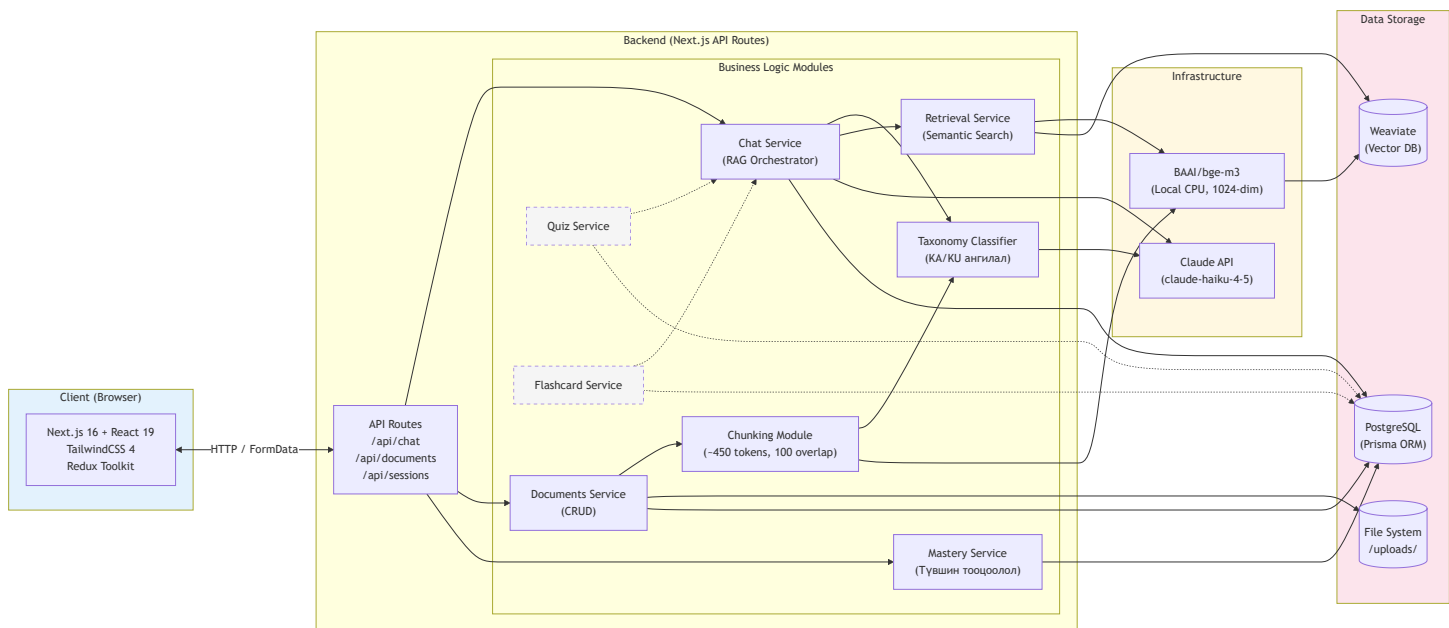






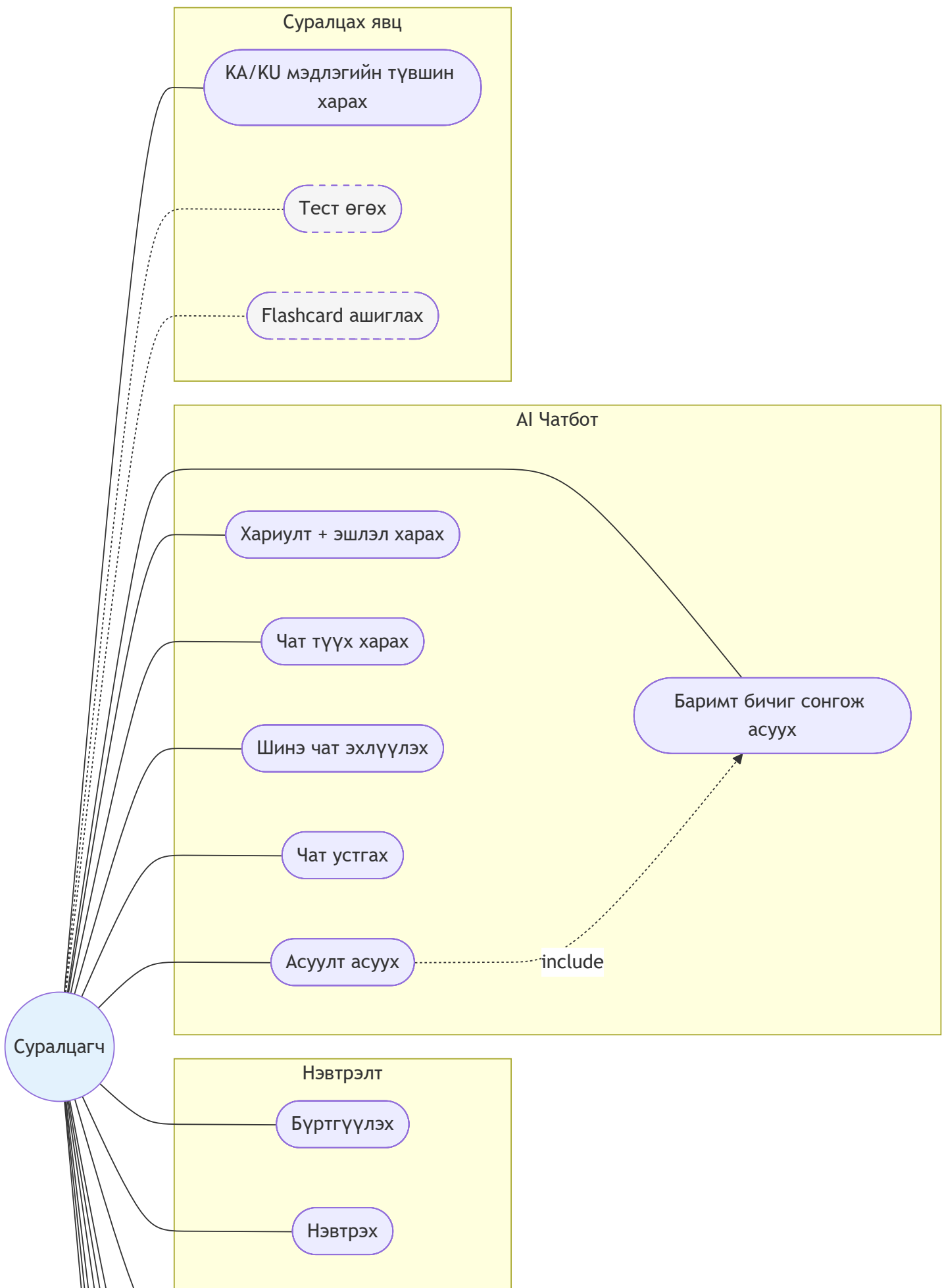


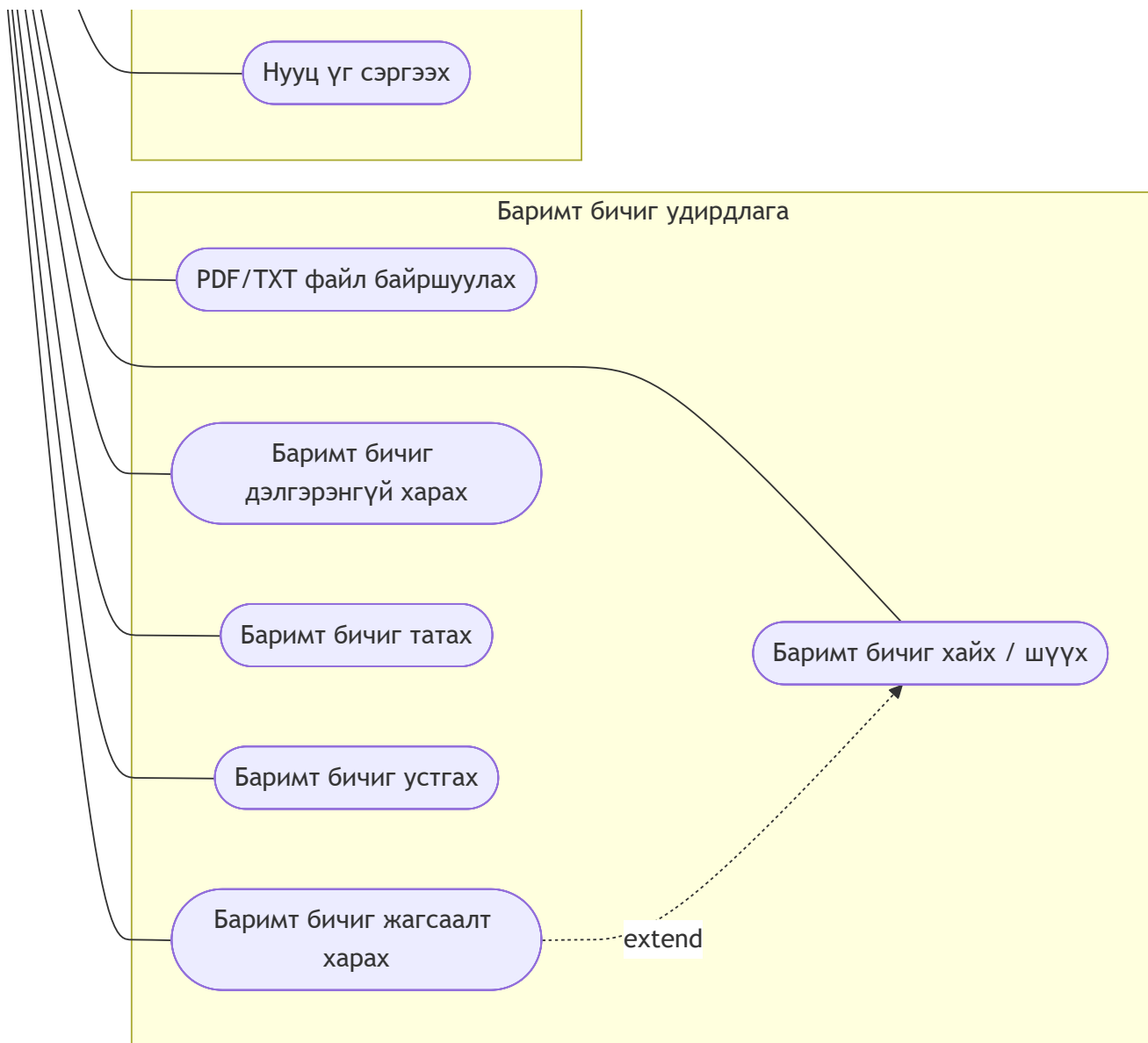
2. Ерөнхий архитектурын диаграмм



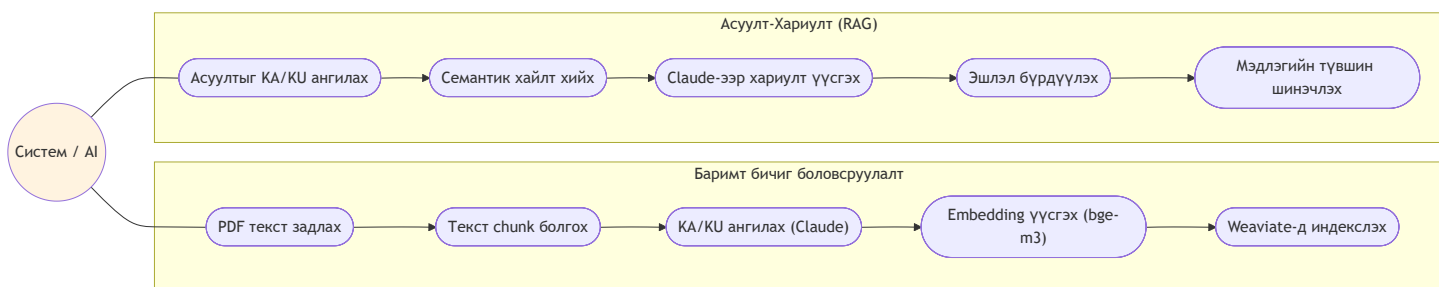
3. Use Case диаграмм

3.1 Суралцагчийн Use Case

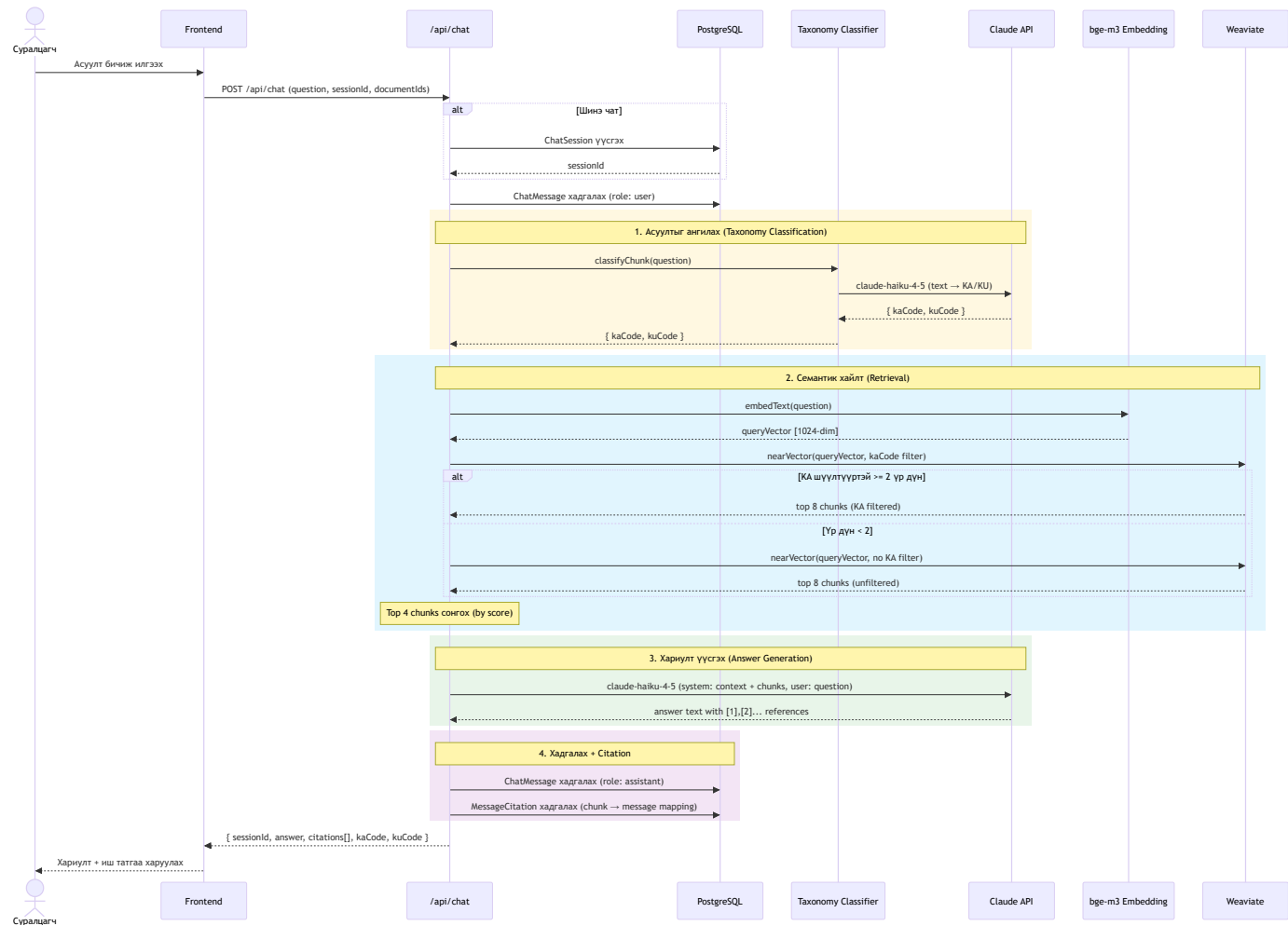




3.2 Системийн автомат Use Case



4. Sequence Diagram — RAG Pipeline (Асуулт-Хариулт)



4.1 RAG Pipeline — Mermaid биш (Markdown Text

Diagram)

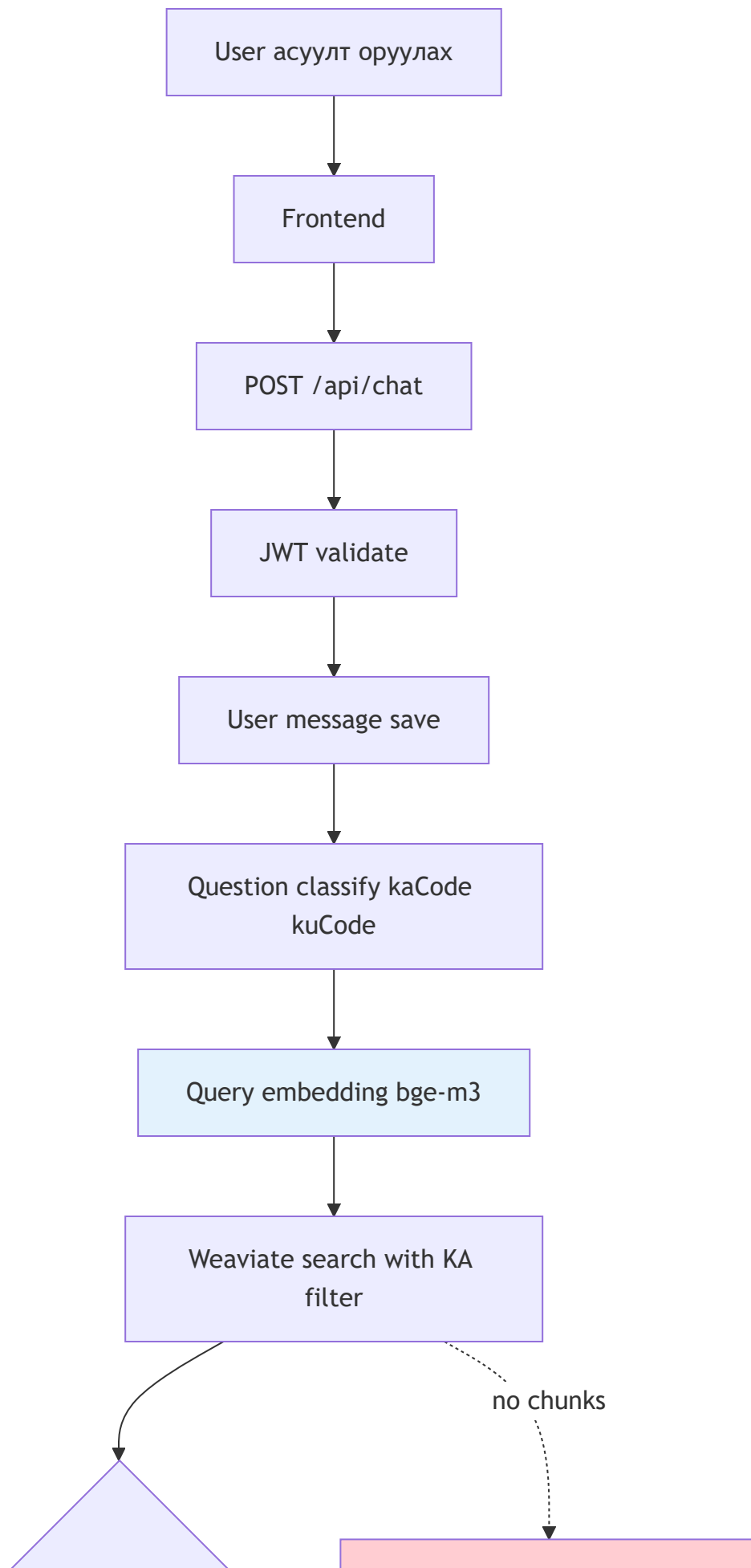
```
[User]
|
| 1) Асуулт оруулах + documentIds сонгох
v
[Frontend]
|
| POST /api/chat (question, sessionId?, documentIds[])
v
[API Route]
|
|-- 2) JWT validate
|-- 3) User message save (PostgreSQL: chat_messages)
|
|-- 4) Question Taxonomy Classification
|   API -> Taxonomy Classifier -> Claude
|   result: kaCode, kuCode
|
|-- 5) Retrieval
|   5.1 Query embedding (bge-m3)
|   5.2 Weaviate nearVector + filters(userId, documentIds, kaCode)
|   5.3 Хэрэв KA-filter үр дүн < 2 бол fallback (kaCode-гүй)
|   5.4 Top-8 -> Top-4 chunks selection
|
|-- 6) Answer Generation
|   Claude-д context(top-4 chunks) + user question илгээх
|   result: grounded answer
|
|-- 7) Citation Mapping
|   assistant message save
|   message_citations save (chunkId, pageNumber, score...)
|
|-- 8) Optional Mastery Update
|   chat evidence record (fail бол non-blocking)
|
v
[API Response]
{ sessionId, answer, citations[], kaCode, kuCode }
|
v
[Frontend Render]
```

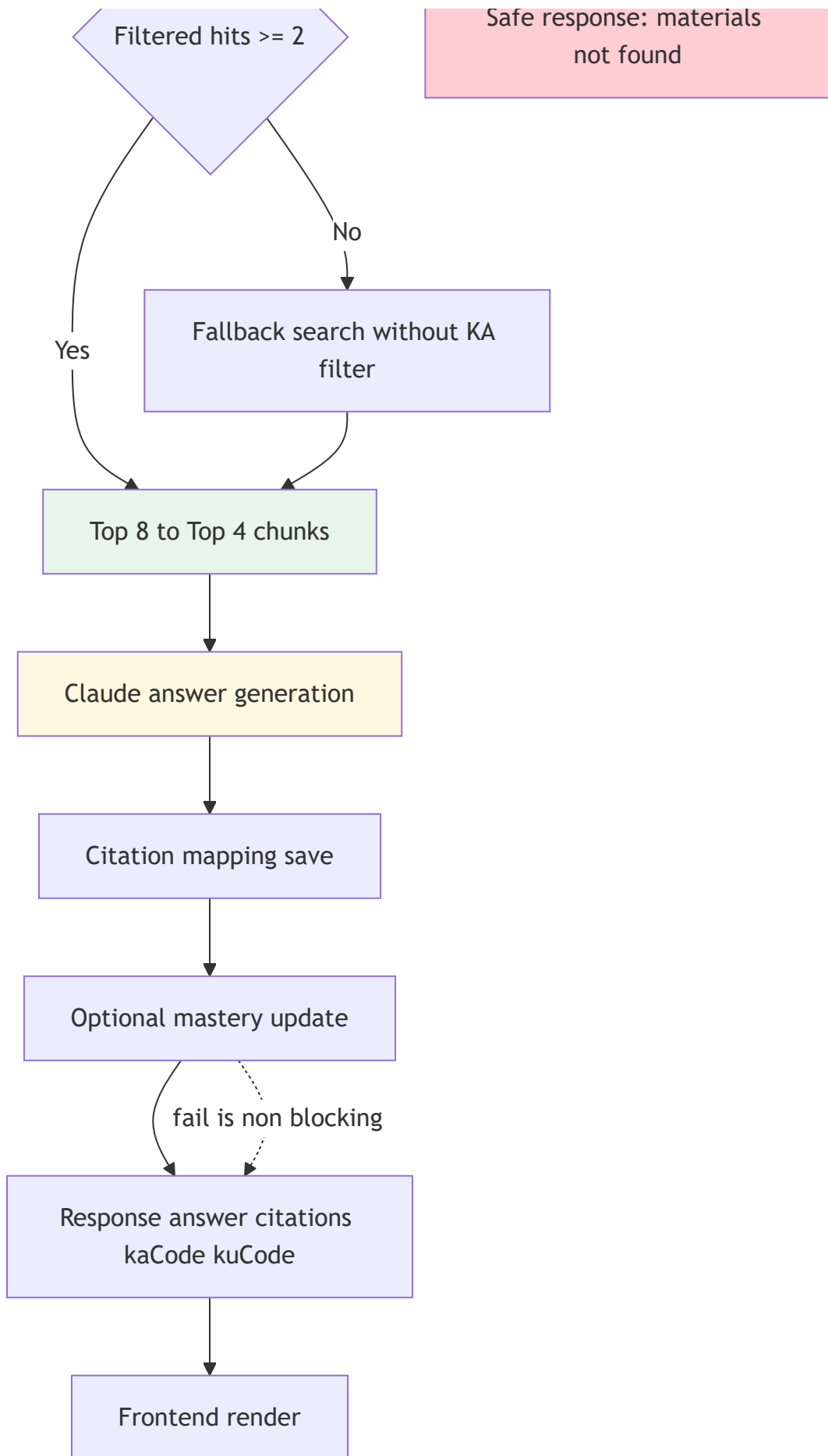
- Хариулт
- Эшлэл
- Follow-up question loop

Error/Edge Paths:

- Хэрэв retrieval хоосон бол: "материалаас олдсонгүй" safe response
- Хэрэв mastery update алдаа гарвал: хариултыг буцаах хэвээр

4.2 RAG Pipeline — Mermaid Diagram





5. ER Diagram — Өгөгдлийн сангийн бүтэц

